

Vivian Zhao

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EDUCATION

Rutgers University - New Brunswick

New Brunswick, NJ

B.A. in Computer Science, B.A. in Data Science

May 2026

- **Awards/Honors:** Honors College, Phi Beta Kappa Society, Dean's List
- **Relevant Coursework:** Introduction to Artificial Intelligence, Data Structures, Computer Architecture, Design and Analysis of Computer Algorithms, Introduction to Data Science, Regression Methods

WORK EXPERIENCE

DIMACS

Piscataway, NJ

Machine Learning Extern

May 2025 - Aug 2025

- Designed binary predictive models using interpretable algorithms (SPLIT, RESPLIT) and benchmarked performance against black-box algorithms (XGBoost) on tabular datasets
- Applied data preprocessing techniques including class balancing (SMOTE) and feature selection to improve model quality
- Analyzed model accuracy vs. interpretability trade-offs and presented weekly findings to DIMACS mentors

Synchrotron Radiation for Biomedicine (STROBE) Laboratory

Grenoble, France

Computer Programming Intern

May 2024 - July 2024

- Trained an AI classification model to autonomously segment and score histological lung scans on a 4-tier inflammation scale
- Defined training data for inflammation severity and validated model accuracy through blind tests across experimental groups
- Conducted data analysis in Excel and statistical correlations in R to quantify tissue inflammation indices

Rutgers Office of Information Technology

New Brunswick, NJ

IT Consultant

Apr 2023 - May 2026

- Supported a technologically efficient environment for 40,000+ patrons through in-person and online consultation
- Leveraged 5+ university software platforms to assist students with network, program, and virus troubleshooting
- Monitored and oversaw university computer labs and printing services 9+ hours per week

PROJECTS

Learning-Based Heuristics for A* Search

Nov 2025 - Dec 2025

Tools: Python, PyTorch, NumPy, Scikit-learn, Matplotlib

- Trained neural network-based heuristics to approximate cost-to-goal for A* search, improving efficiency over classical admissible heuristics
- Generated supervised training data from optimal search runs and evaluated performance via runtime and node expansions

LEADERSHIP & ACTIVITIES

Rutgers Club Tennis

May 2025 - May 2026

President

- Served as liaison to the Sports Club Council and managed administrative responsibilities including executive board oversight, court contract negotiation, and monthly council meetings
- Directed team operations such as practice scheduling, tournament logistics, roster management, and compliance with regulations

Rutgers Creative Expressions Publication

Feb 2023 - Apr 2023

UI/UX Fellow

- Collaborated in a 10-week program to conceptualize and prototype a mobile social app for rating and reviewing music
- Produced user flows, low fidelity wireframes, and high-fidelity prototypes with a focus on UX research and visual design

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, R, C, JavaScript, HTML, CSS, MATLAB, OCaml

Tools: VS Code, Git, PyTorch, Scikit-learn, Jupyter Notebook, RStudio, Tableau, React, Figma, MySQL Workbench, Eclipse, Bootstrap, jQuery, ServiceNow, ImageJ, LaTeX, Adobe Creative Suite